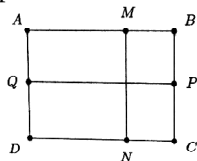


AMTI (NMTC) - 2014
GAUSS CONTEST - PRIMARY LEVEL

PART - A

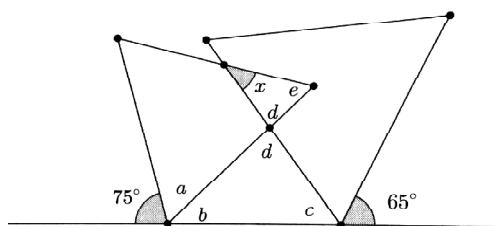
1. Consider the numbers 2, 3, 4, 5. Form two digit numbers of different digits using these numbers. How many of them are odd ?
(A) 4 (B) 5 (C) 6 (D) 7
2. A is the sum of all even three digit numbers in which all the three digits are equal. B is the sum of all odd three digit numbers in which all the digits are equal. The value of $\frac{B}{A}$ is
(A) $\frac{5}{4}$ (B) $\frac{4}{3}$ (C) $\frac{6}{5}$ (D) $\frac{7}{6}$
3. 21 rose plants, 42 sunflower plants and 56 dalia plants have to be planted in rows such that each row contains the same number of plants of one variety only. The minimum number of rows in which the above plants may be planted is
(A) 3 (B) 15 (C) 17 (D) 21
4. The length and breadth of a square are increased by 30% and 20% respectively. The area of the rectangle so formed exceeds the area of the square by
(A) 25 % (B) 50 % (C) 60 % (D) 56 %
5. ABCD is a rectangular play ground in which AB = 40 m and BC = 30 m. The Physical director of the school gave punishment to two students Samrud and Saket. Samrud has to start from A and go round the play ground along ABCDA twice.
Saket has to start from A go along AM, MN, NC, CP, PQ and to A two times. MN is perpendicular to CD and PQ perpendicular to DA. Then



- (A) Samrud covers more distance than Saket
(B) Saket covers more distance than Samrud
(C) Samrud and Saket cover equal distances
(D) Saket covers exactly twice the distance Samrud covers
6. B has 5 Rupees more than C. A has 14 Rupees more than B. Which transaction makes equal money to all the three.
(A) A gives 6 Rs. to B and B gives 3 Rs. to C
(B) A gives 3 Rs. to B and C receives 6 Rs. from A
(C) A gives 2 Rs. to C and B gives 5 Rs. to C
(D) A gives 8 Rs. to C and B receives 3 Rs. from A
7. A natural number a is multiplied by 11 and 33 is added to it. It is divided by 9 and the remainder is zero. The smallest such natural number is
(A) 15 (B) 3 (C) 6 (D) 1
8. n is a natural number such that $(n + 1)(n + 2)(n + 3) = 201320132013$. Then
(A) n is a two digit number (B) n is a three digit number
(C) No such n exists (D) n is a three digits number ending with 3
9. What fraction of the rectangle is shaded ?
(A) $\frac{1}{2}$ (B) $\frac{17}{30}$ (C) $\frac{5}{8}$ (D) $\frac{8}{15}$
10. The first and the third digit of a five digit number d6d41 are the same. If the number is divisible by 9, the sum of its digits is
(A) 18 (B) 36 (C) 25 (D) 27

PART - B

11. Good Shepherd high school has 1584 students. $\frac{4}{9}$ of the students are girls. The number of boys more than the girls in the school is _____.
12. There is a lengthy rope. Ram cuts one third of the rope. From the remaining Rahim cuts one fifth. The total length of the rope both cut is 861 metres. The length of the original uncut rope is _____.
13. Square papers of black and white are arranged as shown :
 ■ ■ □ □ ■ □ ■ ■ □ □ ■ □ ■ ■ □
 There are totally 80 squares. The number of white squares is _____.
14. A company promised Govind Rs. 21,000 and a gift for working 2 years. But Govind left the job in 16 months and got Rs. 12,500 and the gift as compensation. The gift is worth of Rs. _____.
15. Two equilateral triangles overlap as in the figure. The value of the angle x is _____.



16. x and y are the digits of a two digit number xy. x is greater than y by 3. When this two digit number is divided by the sum of its digits the quotient is 7 and the remainder is 3. The sum of the digits of the two digit number is _____.
17. If the square roots of the natural numbers 1 to 200 are written down, the number of whole numbers among them is _____.
18. Two ants start at A and walk at the same speed, one along the square and the other along the rectangle. The minimum distance (in cm) any one must cover before they meet again is _____.
19. When 26 is divided by a positive integer N, the remainder is 2. The sum of all possible values of N is _____.
20. If $\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{4}\right)\left(1 + \frac{1}{6}\right)\left(1 + \frac{1}{8}\right)\left(1 + \frac{1}{10}\right) \times \left(1 - \frac{1}{3}\right)\left(1 - \frac{1}{5}\right)\left(1 - \frac{1}{7}\right)\left(1 - \frac{1}{9}\right) = 1 + \frac{1}{n}$ then the value of n is _____.

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